

Platform Use vs Mood:

A Comparative Analysis of Platform-Specific Social Media Effects

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DSA210 – Term Project

1. Motivation

Social media is often treated as a single behavior, yet different platforms expose users to different types of content and interactions. Based on personal observation, I noticed that platforms such as Instagram, Twitter/X, TikTok, and YouTube affect my mood, focus, and motivation in distinct ways. This project was motivated by the desire to systematically analyze these platform-specific effects using data science methods and self-collected daily data.

2. Data Source

The dataset used in this project is self-collected over a period of 21 days. Daily screen time data for Instagram, Twitter/X, TikTok, and YouTube was recorded manually from the phone's screen time dashboard. In addition, self-reported well-being scores were recorded at the end of each day.

The well-being variables include anxiety, social comparison, productivity, and motivation, each measured on a 1–10 scale. Each row in the dataset represents a single day, combining platform usage metrics with emotional and behavioral outcomes.

3. Data Analysis

The analysis was conducted in three main stages. First, exploratory data analysis was performed to examine distributions, trends, and correlations between platform usage and well-being variables. Visualizations and summary statistics were used to identify initial patterns.

Second, hypothesis testing was applied to examine whether relationships between platform usage and well-being measures were statistically significant. These tests helped evaluate predefined research questions regarding anxiety, social comparison, and productivity.

Finally, regression modeling was used to quantify the strength and direction of platform-specific effects. Simple and multiple linear regression models were applied to examine how daily usage across platforms predicts well-being outcomes. Regression coefficients were compared to identify the strongest predictors.

4. Findings

The results suggest that different social media platforms are associated with distinct well-being outcomes. Twitter/X usage shows a stronger association with higher anxiety levels, potentially due to exposure to fast-paced and negative content. Instagram and TikTok usage are more strongly related to social comparison, likely driven by appearance-focused and curated content.

In contrast, YouTube usage demonstrates a weaker or sometimes positive relationship with productivity, particularly when the platform is used for long-form or educational content. Overall, the findings support the idea that social media should not be treated as a homogeneous activity.

5. Limitations and Future Work

This project has several limitations. The dataset is small and self-collected, which limits generalizability and introduces potential self-report bias. Additionally, the analysis is observational and does not imply causal relationships.

Future work could include collecting data over a longer period, involving multiple users, and incorporating content-type information such as educational versus entertainment usage. Time-lagged analyses could also be explored to examine delayed effects of platform usage on well-being.